TandemSDR v1.1

TandemSDR is a self-contained SDR remote control that runs entirely in a browser with no installation required. It links to SDRplay's SDRconnect software via WebSocket and provides a full receiver control interface including spectrum display, waterfall, audio output, signal metering (dBm and SNR), and signal history charting.

Features

- **WebSocket connection to SDRconnect**
    - Connects to SDRconnect running locally or on a network.
    - Default connection: 127.0.0.1 port 5454 (both configurable). A hostname like *labcomputer* can also be used if SDRconnect is Windows admin mode.
    - IP addresses and names are saved in browser local storage and survive page reloads. Accessible from dropdown menu.
    - Will connect over the WAN if router has port 5454 forwarding enabled.
    - Connection status indicator shows Connected, Connecting, or Disconnected.
    - On connect, all current device state is retrieved automatically from SDRconnect.
- **Frequency Control**
    - Separate VFO and Center frequency inputs in MHz.
    - Hold-to-repeat step buttons and mouse scroll wheel for VFO tuning. Step buttons accelerate the longer they are held.
    - Selectable step size: 10 Hz, 100 Hz, 500 Hz, 1 kHz, 2.5 kHz, 5 kHz, 10 kHz, 12.5 kHz, 25 kHz, 100 kHz.
    - Hold-to-repeat step buttons for Center frequency using the selected step size.
    - VFO → Center button to sync the center to the current VFO frequency. When sample rate is above 2 MSPS a 10 kHz offset is applied to keep the VFO out of the DC centre of the spectrum.
- **Device Controls**
    - Sample rate selector: 62.5 kSPS to 10 MSPS.
    - Device selector — populated dynamically from SDRconnect.
    - Antenna selector — populated dynamically from SDRconnect.
    - RF Gain slider with overload warning indicator.
- **Demodulation**

    - Modes: AM, SAM (Synchronous AM), USB, LSB, CW, NFM, WFM.
    - Each mode sets a default filter bandwidth on selection.
    - Filter bandwidth input with hold-to-repeat step buttons that accelerate the longer they are held. Maximum bandwidth is enforced per mode.
    - WFM mode requires a minimum sample rate of 250 kSPS; the 62.5 and 125 kSPS options are disabled when WFM is selected. If either of those rates is active when WFM is selected, the sample rate automatically jumps to 2 MSPS.
    - RDS on/off toggle with PI, PTY, and PS display (WFM mode).
    - Stereo on/off toggle with stereo indicator.
- **Audio**
    - Audio in SDRconnect is muted after TandemSDR is connected.
    - Audio streamed from SDRconnect and decoded in the browser via the Web Audio API.
    - Volume control slider with mute toggle button.
    - Noise Reduction on/off toggle with strength slider.
    - Audio AGC on/off toggle with threshold slider.
    - Squelch on/off toggle with level slider.
- **S-Meter**
    - Segmented bar display calibrated to IARU standard.
    - HF mode (below 30 MHz): S9 = −73 dBm, 6 dB per S-unit.
    - VHF mode (30 MHz and above): S9 = −93 dBm, 6 dB per S-unit.
    - Segments S1–S9 in green, +10 through +60 dB over S9 in red.
    - Automatically switches HF/VHF calibration based on current frequency.
- **Signal readouts**
    - Live Signal Power display in dBm.
    - Live Signal SNR display in dB.
    - Live Frequency display.
- **Spectrum Display**
    - FFT spectrum analyzer. (512 points)
    - Adjustable Max (reference) and Min (base) level inputs.
    - Frame averaging: None, 2, 4, 8, 16 frames.

    - Zoom: ×1, ×2, ×4 — cycled via button or keyboard.
    - Frequency snap grid: 1 kHz, 5 kHz, 9 kHz, 10 kHz, 25 kHz, 100 kHz.
    - Crosshair follows mouse with frequency readout.
    - Left-click sets VFO frequency at cursor position.
    - Right-click sets Center frequency at cursor and syncs VFO.
    - Drag on the frequency label row to pan the center frequency.
    - Scroll wheel steps the VFO frequency.
- **Waterfall Display**
    - Scrolling waterfall below the spectrum.
    - Optional frame averaging (follows spectrum averaging setting).
    - Speed selector: Slow, Medium, Fast.
    - Same mouse controls as the spectrum (left-click, right-click, scroll wheel).
- **Signal History Chart**
    - Time-series chart of Signal Power (dBm) or SNR over time.
    - Selectable capture interval: 500 ms, 1 sec, 5 sec, 10 sec, 30 sec, 60 sec.
    - Start/Stop/Clear capture control.
    - Session min and max values with timestamps displayed on the chart.
    - Scrollable history slider to review up to 10× the visible window of captured data. Automatically snaps back to live after 20 seconds of inactivity.
    - Crosshair mouseover shows exact value and timestamp when stopped or in history mode.
    - **Record** — opens a native file save dialog and streams data directly to a CSV file in real time as samples are captured.
    - **Export** — saves the current in-memory buffer to a CSV file via a native file save dialog. CSV includes frequency, mode, capture interval, and timestamped power and SNR columns.
- **Memory**
    - 10 memory slots for saving and recalling receiver settings.
    - Each slot stores VFO frequency, center frequency, mode, filter bandwidth, sample rate, antenna, RF gain, step size, volume, and spectrum display settings (reference level, base level, zoom, averaging, snap).
    - Slots are saved in browser local storage and survive page reloads.

    - Individual slots can be cleared from the memory modal.
- **Band Selector**
    - Quick-access modal for jumping to common bands with one click.
    - Sets center frequency, sample rate, mode, snap, and step for the selected band. Also sets RF gain to maximum and turns squelch off.
    - Covers Amateur Radio (160 m through 33 cm), Shortwave Broadcast (120 m through 11 m), and Broadcast (AM and FM).
- **Keyboard Shortcuts** (hotkeys)

Key	**Action**
← →	Step VFO by selected step size
↑ ↓	Step Center by Snap value
PgUp PgDn	Jump Center one full span width
=	Copy VFO → Center
A	AM mode
S	SAM mode
U	USB mode
L	LSB mode
C	CW mode
N	NFM mode
W	WFM mode
V	Toggle audio mute
Q	Toggle squelch
D	Toggle Spectrum Display
F	Toggle WaterFall (enables Spectrum first if hidden)
Z	Cycle spectrum zoom (×1 → ×2 → ×4)
+	Increase sample rate
-	Decrease sample rate
M	Save memory slot
R	Recall memory slot
B	Open Band Selector
H or ?	Open Help
Esc	Close any open modal

- **Built-in Help**
  - Help modal accessible via the ? button or H/? keyboard shortcut.
  - Covers all mouse controls for spectrum and waterfall.
  - Full keyboard shortcut reference.

Requirements

- No installation and no internet connection required. Open the HTML file directly in your browser
- **SDRplay SDRconnect** must be running with WebSocket access enabled on port 5454
- Compatible with any SDRplay RSP device supported by SDRconnect.
- Works in any modern browser. The **Record** and **Export** features require a Chromium-based browser (Google Chrome, Microsoft Edge, Brave, or Opera) as they depend on the File System Access API, which is not supported in Firefox or Safari.

Typical Workflow

1. Start **SDRconnect** and ensure it is running with WebSocket enabled. See image below
2. Open **TandemSDR** by double-clicking the HTML file in your browser.
3. Enter the IP address and port of the SDRconnect instance (defaults to 127.0.0.1:5454).
4. Click **Connect**. TandemSDR will retrieve all current device settings automatically.
5. Use the **VFO** input or click on the **Spectrum** to tune to a frequency.
6. Select a demodulation **Mode** and adjust **Filter BW** as needed.
7. Adjust **RF Gain**, **Volume**, **Noise Reduction**, and **Squelch** as desired.
8. Click **Show** under Spectrum Display to enable the spectrum and waterfall.
9. Use **Signal History** to capture and export signal power or SNR over time.
10. Hotkeys (listed above) allow quick changes to features and parameters.

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**** Enable WebSocket in SDRconnect before using program.**